6 Everyday use and help

FULL

06. Shopping list

Quick version

No time to read it all? This is the short version. The full guide is below.

- What to buy for the projects, with rough prices.
- Buy things only when you reach the project that needs them. Projects 4, 6 and 7 need parts.
- The whole ladder costs about $80 to $150, spread over weeks. Prices change: check before you buy.
- For Project 7, the valve must be **normally closed**: water is OFF when the power is off.
- Do not buy anything that says "mains", "110V", "220V" or "AC" for these projects.
- Not sure a part matches the list? Ask Gemini with prompt 6 (Check my plan) before buying.
- Buy from stores that list returns.

You need **nothing** to do guides 01 to 03 and Projects 1 to 3, 5 and the Home Assistant part of Project 3 beyond the laptop. Buy things only when you reach the project that needs them.

Prices are rough ranges and change. Check the current price before you buy.

For	Part	Rough price	Notes
Guide 01	USB stick, 16 GB or larger	$5 to $10	You may have one. It gets erased.
Guide 01, a Pi instead of a laptop	**Raspberry Pi 5**, 8 GB of memory	Check the official shops below	The 4 GB one works too (it was tested), 8 GB gives more room.
Same	The **official Raspberry Pi 27 W USB-C power supply**	Same	A weaker phone charger makes a Pi 5 unstable.
Same	The official **Active Cooler** (a fan), or the official **Case** with its fan	Same	Needed: without a fan, the Pi slowed itself down at 84.5 °C while building the garden (guide 01).
Same	An **SSD** with the **M.2 HAT+**, or a USB 3 SSD; or a 128 GB+ microSD card	Same	An SSD is faster and lasts much longer than a memory card (the garden writes every day).

Raspberry Pi's official products and their list of approved shops: https://www.raspberrypi.com/products/. | Project 4 | A Govee light that supports **LAN control** | $15 to $40 | Check the model's page or the Govee app for "LAN Control" before buying. Older models only work through the cloud. | | Project 6 | **ESP32 dev board** (ESP32-WROOM-32, with USB) | $6 to $12 | Buy 2; one is a spare for mistakes. | | Project 6 | **Capacitive** soil moisture sensor v1.2 (not "resistive") | $2 to $4 each | Capacitive ones do not rust. Buy 2 to 4. | | Project 6 | USB cable that carries data (not charge-only) | $5 | If the laptop does not see the board, the cable is the usual cause. | | Project 6 | DHT22 or BME280 air sensor (optional) | $4 to $10 | BME280 gives temperature, humidity and pressure. | | Project 7 | 12 V **normally-closed** water solenoid valve, 1/2 inch, garden-hose type, **or** a small 5 V/12 V submersible pump with tubing | $10 to $25 | Normally closed means water is OFF when power is off. This is the safe one. | | Project 7 | **Relay module**, 1 channel, 5 V, opto-isolated | $3 to $6 | It is the switch the ESP32 uses. | | Project 7 | 12 V DC power adapter, 2 A, with a screw-terminal adapter | $8 to $12 | Low-voltage power for the valve. | | Project 7 | **Flyback diode** 1N4007 | $1 | Protects the relay from the valve's electrical kick. | |

Project 7 | Waterproof box (IP65), hose fittings, Teflon tape, drip tubing | $15 to $30 | Electronics and water must be kept apart. | | Project 7 | 2 A fuse and holder for the 12 V line | $3 | A fuse stops a fire if something shorts. | | Any | Breadboard, jumper wires, small screwdriver set | $10 to $15 | A starter kit has all of this. |

Total for the whole ladder: about $80 to $150, spread over weeks.

Rules for buying

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- Do not buy anything that says "mains", "110V", "220V" or "AC" for these projects.
- If a part does not match this list, ask Gemini with prompt 6 (Check my plan) before buying.